

1. Demonstrate that the determinant of any matrix containing a row of zeros is zero.
2. Compute the determinant of the following matrix?

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 2 & 0 & -1 \\ 1 & 4 & 5 \end{bmatrix}$$

3. Demonstrate that each time you swap a row in a square matrix, you are going to flip the sign of the determinant without actually changing the magnitude of the determinant. Consider the matrix in task 2 and swap the second and third rows.
4. For the following two matrices confirm

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

5. Is the following matrix a singular or non-singular matrix?

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$